

Fine-Tuning, Not Windowing, Closes the Gap: 3D CNNs vs. Pretrained 2.5D Classifiers for Multi-Label Chest CT

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Abstract. Chest CT abnormality classification is complicated by patient-level data leakage, limited labelled 3D data, and the choice of intensity windowing. A leakage-free, patient-isolated pipeline for 18-class multi-label classification on a 1,000-volume CT-RATE subset is used to run a controlled comparison of model configuration (3D ResNet50 trained from scratch vs. a 2.5D classifier built on a frozen or fine-tuned ImageNet-pretrained ResNet50 with attention pooling) and HU intensity representation (single fixed window vs. a three-channel lung/soft-tissue/bone window). Under a restricted training regime of 700 volumes, multi-window preprocessing does not improve either configuration, and a frozen 2.5D classifier underperforms the 3D-from-scratch baseline despite having fewer than 1.5% as many trainable parameters. Fine-tuning the multi-window 2.5D backbone achieves the numerically highest AUROC in the study (0.6826) at the cost of clear overfitting past its best-validation-loss checkpoint. Among the classes examined in detail, fine-tuning produces particularly large gains for lymphadenopathy, mosaic attenuation pattern and medical material, while rare classes show poor threshold-dependent performance at the fixed 0.5 decision boundary despite above-chance AUROC. Grad-CAM and attention-weight visualisations illustrate these patterns on representative cases, and the implications and limitations of the study are discussed in relation to model and preprocessing choices under limited labelled 3D CT data. Code for this project is available at https://github.com/alva-phelan/Data-Challenges_RDGY41850.

1 Introduction

Computed tomography (CT) is central to the diagnosis and assessment of thoracic disease, but interpretation of volumetric scans is time-consuming and a single examination may contain multiple co-occurring abnormalities across hundreds of slices. Automated multi-label classification therefore has the potential to support radiologists by rapidly screening CT volumes for clinically relevant findings.

Three-dimensional convolutional neural networks (3D CNNs) provide a natural approach to learning directly from volumetric data [6], but their computational and data requirements can limit performance when labelled datasets are

small. Recent work has described this as *data-regime bias*, showing that pre-trained two-dimensional representations can remain competitive with, or out-perform, 3D models trained from scratch under limited labelled-data regimes [9]. Reliable evaluation is also important because repeated examinations from the same patient can introduce leakage if scans are randomly partitioned across dataset splits, leading to overly optimistic estimates of generalisation [3,14].

Input representation provides a second source of variation: different tissues are best visualised under different Hounsfield Unit (HU) window settings, so a single fixed window may compress complementary lung, soft-tissue, and bone information [11,13].

This project investigates how model configuration and CT intensity representation interact under a restricted labelled-data regime for 18-class chest abnormality classification on a subset of CT-RATE [5]. A randomly-initialised 3D ResNet50 is compared with a 2.5D transfer-learning approach based on an ImageNet-pretrained 2D ResNet50 [6,2], where sampled 2D slice features are combined using attention pooling [7]. This varies initialisation and dimensionality together (Sect. 7). Both configurations are evaluated using identical patient-isolated data splits under single-window and three-channel lung, soft-tissue, and bone representations. The main contributions are: A leakage-free, patient-isolated pipeline for 18-class multi-label classification on a restricted subset of CT-RATE, a controlled comparison between a randomly-initialised 3D ResNet50 and an ImageNet-pretrained 2.5D classifier under a limited-data regime, a model-configuration $\times$ preprocessing comparison of single-window and multi-window HU representations and an evaluation of backbone fine-tuning on the multi-window 2.5D configuration, supported by attention-based and Grad-CAM [12] visualisations of model behaviour.

2 Related Work

2.1 Chest CT Abnormality Classification and CT-RATE

Chest CT classification is inherently multi-label as several abnormalities may coexist within a single examination, creating heterogeneous appearances and substantial class-prevalence differences. CT-RATE [5] provides a large-scale resource for this task, pairing chest CT volumes with 18 binary abnormality labels, and forms the basis of the VLM3D Challenge. This study focuses on supervised classification using these labels under a restricted-data setting, with patient-level splitting used throughout to avoid the leakage risk noted above [3,14].

2.2 3D Models and Pretrained 2D Representations

ResNet-based 3D CNNs have been widely adapted to volumetric medical imaging [6], but their capacity can exceed the available supervision when trained from random initialisation on small labelled datasets, increasing susceptibility to over-fitting [9]. An alternative represents a 3D examination as a collection of 2D slices

processed by a shared pretrained network, aggregated into a volume-level representation. Liu et al. [9] term this comparison *data-regime bias*, showing that pre-trained 2D representations can outperform volumetric models when labelled 3D data are limited. This study examines this hypothesis using approximately 700 training volumes, smaller than [9]’s setting, comparing a 3D ResNet50 trained from scratch against a 2.5D model using an ImageNet-pretrained ResNet50 [6,2] and attention pooling [7], with frozen and fine-tuned backbones to test whether generic pretrained features are sufficient or require CT-domain adaptation.

2.3 HU Windowing for Chest CT

The clinically useful Hounsfield Unit range varies across lung parenchyma, mediastinal soft tissue, and bone, motivating multi-window representations that preserve complementary intensity information across several window settings simultaneously. M3Lung-Sys [11] used multiple CT windows for pulmonary classification, and more recent CT-RATE-derived work has incorporated lung, soft-tissue, and bone windows specifically [13]. This paper compares conventional single-window preprocessing directly against this three-channel lung, soft-tissue, and bone representation, under both the 3D and 2.5D architectures.

3 Method

3.1 Overall Framework

Figure 1 summarises the experimental framework. All models share the same patient-level split and spatial preprocessing before branching along two experimental axes: intensity representation (single-window vs. multi-window) and model configuration (3D ResNet50 trained from scratch vs. a pretrained 2.5D classifier). A frozen/fine-tuned backbone ablation was additionally performed for the 2.5D model.

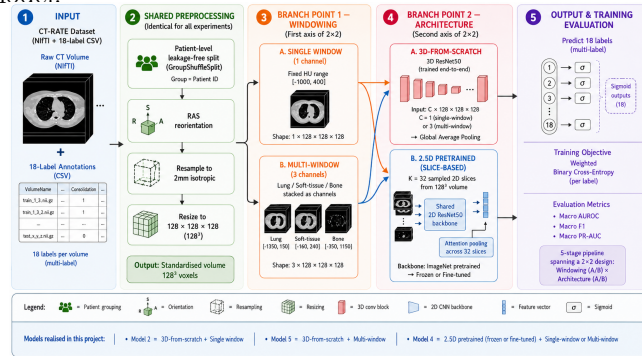


Fig. 1: Overall experimental framework. All models share the same leakage-free patient-level split and spatial preprocessing before comparison across windowing and model configurations.

Table 1: Per-class positive count and prevalence (% of split) across the train/validation/test splits (702/148/150 scans, 366/78/79 patients), showing the two most and two least prevalent classes by test-set frequency. Full 18-class statistics are provided in the project repository.

Class	Train n (%)	Val n (%)	Test n (%)
Lung nodule	326 (46.4%)	69 (46.6%)	88 (58.7%)
Arterial wall calcification	210 (29.9%)	44 (29.7%)	69 (46.0%)
Pericardial effusion	58 (8.3%)	2 (1.4%)	8 (5.3%)
Mosaic attenuation pattern	50 (7.1%)	8 (5.4%)	8 (5.3%)

3.2 Data Processing

A subset of 1,000 CT volumes (702 training, 148 validation, 150 test) from CT-RATE [5] was used for 18-class multi-label classification, with a maximum of two scans per patient to reduce over-representation of individual patients. Volumes were processed using MONAI [1], including reorientation to RAS space, resampling to 2.0 mm isotropic voxel spacing, and resizing to $128 \times 128 \times 128$. Patient-level splits were generated using `GroupShuffleSplit` [10] with 70% training, 15% validation, and 15% test data. Random flips and affine transformations (rotation and translation) were applied during training only.

Single-window input. The baseline representation used fixed HU windowing over $[-1000, 400]$, linearly rescaled to $[0, 1]$.

Multi-window input. Clinically motivated HU windows were independently clipped and rescaled to $[0, 1]$: lung (level -600 , width $1500 \rightarrow [-1350, 150]$), soft tissue (level 40 , width $400 \rightarrow [-160, 240]$), and bone (level 400 , width $1500 \rightarrow [-350, 1150]$) [11,13], stacked to form a three-channel input, replacing the single input channel for the 3D model, and replacing single-window channel replication for the 2.5D model.

2.5D input preparation. For the 2.5D branch, $K = 32$ evenly spaced 2D slices were sampled from each preprocessed volume, resized to 224×224 , and normalised using ImageNet channel statistics.

3.3 Network Architecture

3D ResNet50. The 3D branch used a ResNet50 [6] with 18 output logits, trained end-to-end from random initialisation. The single-window configuration corresponds to Model 2 and the multi-window configuration to Model 5.

2.5D pretrained classifier. Each of the K sampled slices were passed through a shared ImageNet-pretrained 2D ResNet50 backbone f_θ with its final fully connected layer removed, producing slice-level features $\mathbf{h}_k = f_\theta(s_k) \in \mathbb{R}^{2048}$. These features were aggregated using attention pooling [7]:

$$a_k = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{U}\mathbf{h}_k))}{\sum_{j=1}^K \exp(\mathbf{w}^\top \tanh(\mathbf{U}\mathbf{h}_j))}, \quad \mathbf{z} = \sum_{k=1}^K a_k \mathbf{h}_k \quad (1)$$

where $\mathbf{U}$ and $\mathbf{w}$ are learned parameters and a_k is the attention weight assigned to slice k . A linear classification layer maps $\mathbf{z}$ to 18 output logits. The

pretrained backbone was either frozen, with only the attention module and classification head trained, or fine-tuned end-to-end as an ablation.

Loss function. Models used weighted binary cross-entropy with a per-class positive-class weight (PyTorch’s `BCEWithLogitsLoss(pos_weight=...)`) to account for class imbalance:

$$\mathcal{L} = -\frac{1}{C} \sum_{c=1}^C [w_c y_c \log(\hat{y}_c) + (1 - y_c) \log(1 - \hat{y}_c)], \quad w_c = \frac{N_{\text{neg},c}}{N_{\text{pos},c}} \quad (2)$$

where $C = 18$ and the class weight w_c is determined from the ratio of negative to positive training examples for class c .

3.4 Training Strategy

Models were trained using Adam [8] with a learning rate of 1×10^{-4} , batch size 4, and a maximum of 30 epochs. The checkpoint with the lowest validation loss was retained for evaluation. For the frozen 2.5D configuration, only the attention-pooling and classification-head parameters were optimised, while the full backbone was trainable in the fine-tuned configuration.

Hardware: single NVIDIA L40S (48GB VRAM). Models 4 and 5 completed their full 30-epoch schedule, with wall-clock training time ranging from 8h37m (Model 4, frozen backbone, multi-window) to 10h28m (Model 5, 3D-from-scratch, multi-window). Model 2 was terminated after ~ 24 h at the cluster’s wall-time limit and evaluated from its best checkpoint-by-validation-loss (see limitations, Sect. 7).

4 Experiments

4.1 Experimental Setup and Evaluation Metrics

Experiments used the 1,000-volume CT-RATE subset and fixed patient-level split described in Sect. 3.2, following the 2×2 model-configuration $\times$ windowing design of Fig. 1 plus a fine-tuned 2.5D multi-window variant as a backbone-adaptation ablation. Performance was evaluated on the fixed test set using macro-averaged AUROC, F1-score, and PR-AUC (average precision) across the 18 abnormality classes. F1-score used a pre-specified threshold of 0.5. The official VLM3D Task 2 ranking additionally reports CRG [4] and statistical comparisons across submissions. CRG could not be reproduced locally (see limitations, Sect. 7).

5 Results

Two findings stand out. First, multi-window preprocessing did not improve either configuration at this data scale. It reduces 3D-from-scratch AUROC by

Table 2: Overall test performance across the 2×2 model configuration $\times$ windowing design. Single-window rows reuse Models 2/4’s original windowing; multi-window rows use the lung/soft-tissue/bone scheme.

Model	Windowing	Trainable Params	Test AUROC	Test F1	Test PR-AUC
3D-from-scratch (Model 2)	Single window	46.2M	0.6746	0.3702	0.3871
3D-from-scratch (Model 5)	Multi-window	46.2M	0.6336	0.3739	0.3322
2.5D frozen backbone (Model 4)	Single window	0.56M	0.6347	0.3306	0.3399
2.5D frozen backbone (Model 4)	Multi-window	0.56M	0.6316	0.3690	0.3503

Table 3: Computational complexity by configuration. Total/trainable parameters, FLOPs (via `fvcore`), and forward-pass inference latency (batch size 1, single forward pass, same GPU, mean of 20 timed repeats after 5 discarded warm-up passes, with `torch.cuda.synchronize()` called before and after timing). Model 4’s single and multi-window arms are architecturally identical (only pixel content differs), so this table reports one row per backbone state rather than per windowing scheme.

Configuration	Total Params	Trainable Params	GFLOPs	ms/scan
3D-from-scratch, single-window (Model 2)	46.20M	46.20M	325.68	85.6
3D-from-scratch, multi-window (Model 5)	46.24M	46.24M	417.76	96.3
2.5D, frozen backbone (Model 4)	24.07M	0.56M	131.52	30.6
2.5D, fine-tuned backbone (Model 4)	24.07M	24.07M	131.52	30.6

4.1 points (0.6746 $\rightarrow$ 0.6336) and leaves the frozen 2.5D model essentially unchanged (-0.3 points), complicating the motivation drawn from [11,13]. Second, the frozen 2.5D classifier underperforms the 3D-from-scratch baseline at matched (single) windowing (0.6347 vs. 0.6746), which runs counter to a naive reading of the data-regime bias claim in [9], despite having fewer than 1.2% as many trainable parameters (0.56M vs. Model 2’s 46.2M). The fine-tuned variant (Table 5) changes this picture (Sect. 6).

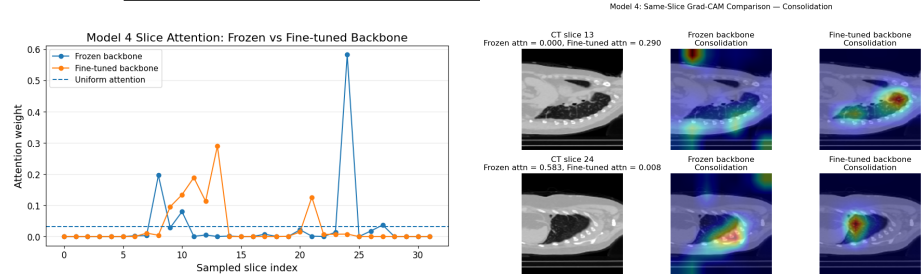
This efficiency advantage extends to inference cost. The 2.5D classifier needs roughly 2.5-3 $\times$ fewer FLOPs (131.52 vs. 325.68–417.76 GFLOPs, Table 3) and runs 2.8-3.1 $\times$ faster per scan (30.6ms vs. 85.6–96.3ms) than either 3D configuration, practically relevant given the VLM3D Challenge’s fully-automatic, containerised inference requirement. Multi-window preprocessing also increases 3D inference cost by 28% while simultaneously reducing AUROC, an unfavourable trade-off in this experiment.

6 Ablation Study

Unfreezing the backbone achieves the numerically highest AUROC in the paper (0.6826, Table 5), exceeding the 3D-from-scratch baseline (0.6746), consistent with the ImageNet-to-CT domain gap limiting the frozen backbone’s performance. With 24.07M trainable parameters but only 702 training volumes, the fine-tuned model had substantial capacity relative to the available supervision, increasing its susceptibility to overfitting. The reported checkpoint (selected exclusively by validation loss) is epoch 7/30 (val. loss 0.9232, val. AUROC 0.7788, val. F1 0.4105). By epoch 30/30, val. loss had risen to 2.7466 while train loss fell to 0.1047, a clear divergence. Validation AUROC itself kept rising over the

Table 4: Per-class test AUROC for classes discussed in Sect. 7, across four of the study’s five configurations (the 2×2 design plus the fine-tuned ablation); the single-window 3D-from-scratch configuration (Model 2) is omitted here. Bold marks the best AUROC per row. Full 18-class, five-configuration results are provided in the project repository.

Class	3D mw (Model 5)	2.5D sw frozen	2.5D mw frozen	2.5D mw fine-tuned
Medical material	0.776	0.693	0.684	0.804
Lymphadenopathy	0.551	0.553	0.515	0.714
Mosaic attenuation pattern	0.626	0.677	0.633	0.788
Pleural effusion	0.894	0.859	0.910	0.944
Peribronchial thickening	0.367	0.362	0.359	0.389
Interlobular septal thickening	0.649	0.649	0.634	0.623
Macro average (all 18 classes)	0.634	0.635	0.632	0.683



(a) Per-slice attention weight, frozen vs. fine-tuned backbone. (b) Illustrative Grad-CAM comparison for Consolidation.

Fig. 2: Attention and Grad-CAM evidence for one representative test case: (a) attention weights across all 32 sampled slices; (b) approximate Grad-CAM localisation under the frozen and fine-tuned Model 4 backbone (illustrative only – attention and Grad-CAM use different slicing axes, see limitations).

first several checkpoints (0.7145→0.7788, epochs 1-7) even as validation loss began climbing soon after, indicating that ranking performance and probability-based validation loss diverged during training. Calibration was not directly evaluated. A mean/max-pooling counterpart under fine-tuning is left to future work (Sect. 8).

7 Discussion

The per-class breakdown (Table 4) shows the fine-tuning gain is not uniform: largest on Lymphadenopathy (0.515→0.714), Mosaic attenuation pattern (0.633→0.788) and Medical material (0.684→0.804), possibly reflecting adaptation to CT-specific features, though the experiment does not establish the cause. Peribronchial

Table 5: Ablation over Model 4 design choices (multi-window input, test set, macro-averaged). Frozen/Attention row repeats Table 2 for reference.

Backbone	Pooling	Trainable Params	Test AUROC	Test F1	Test PR-AUC
Frozen	Attention	0.56M	0.6316	0.3690	0.3503
Fine-tuned	Attention	24.07M	0.6826	0.3800	0.4014

and Interlobular septal thickening are exceptions, both have $F1 = 0.000$ despite above-chance AUROC, given their identical low test-set prevalence (10/150 scans, 6.7% each), likely a threshold rather than discrimination issue (calibration untested). PR-AUC or per-class thresholds would be more informative for rare classes. Peribronchial thickening is hardest under every configuration (AUROC 0.36-0.39), Pleural effusion is the opposite, best-classified throughout (0.86-0.94), improving with both interventions.

Fig. 2a/2b illustrate this on one representative case: frozen-backbone attention spikes on slice 24 (weight 0.583) with near-uniform weighting elsewhere, while fine-tuned attention concentrates over slices 9-13, more consistent with genuine localisation than an isolated outlier. Grad-CAM shows a related pattern (slice-13 weight 0.000→0.290 with sharper localisation), though attention and Grad-CAM use different slicing axes, so both are illustrative rather than exact confirmation of each other.

These findings should be read against several limitations. The 1,000-scan subset (702/148/150) is small relative to full CT-RATE, and the fine-tuned model’s val/test AUROC gap (0.7788 vs. 0.6826) is plausibly inflated variance from split size. Results reflect a single seed and split. The split is patient-isolated but not multi-label stratified. Table 1 shows a corresponding prevalence shift (e.g. Lung nodule, 46.4%→58.7% train-to-test). Architecture and pretraining were varied together (random initialisation vs. ImageNet), so differences cannot be attributed exclusively to dimensionality. Grad-CAM (Fig. 2b) uses `dim=2` while Model 4 samples slices along `dim=1`, so Fig. 2 is representative rather than exactly slice-matched. Our local evaluation also omits CRG [4], the official challenge baseline, and VLM3D’s statistical comparisons, so these results are an internal comparison rather than a challenge reproduction. Finally, Model 2’s checkpoint reflects the best validation loss before the cluster’s wall-time limit rather than a completed 30-epoch run, though its test AUROC (0.6746) already reflected a previously-validated improvement over the leakage-affected baseline.

8 Conclusion

This paper presented a leakage-free, patient-isolated pipeline for 18-class chest classification on CT-RATE, running a 2×2 comparison of model configuration (3D-from-scratch vs. 2.5D pretrained) and HU windowing, plus fine-tuning on the multi-window 2.5D model. Multi-window preprocessing improved neither configuration. A frozen 2.5D classifier underperformed the 3D baseline despite under 1.5% as many parameters, but fine-tuning gave the numerically highest AUROC (0.6826) at the cost of overfitting past epoch 7/30. Since architecture and pretraining were varied together, this reflects model configurations, not a controlled ablation. Among the classes examined in detail, fine-tuning produced particularly large gains for lymphadenopathy, mosaic attenuation pattern and medical material, possibly reflecting adaptation to CT-specific features.

Future work includes full-CT-RATE validation with bootstrap CIs, independently controlling architecture/pretraining, per-class threshold calibration, a pooling ablation under fine-tuning, and medical-domain pretraining.

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